

WASP-2b

R-1682

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_130919 · created 2026-08-02T11:53:04+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456554.8066 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.043
Transit depth (Rp/Rs)^2	1.93 +/- 1.19 [%]
Orbital Inclination (inc)	85.73 +/- 2.77
Airmass coefficient 1 (a1)	1.007 +/- 0.033
Airmass coefficient 2 (a2)	-0.0026 +/- 0.0056
Scatter in the residuals of the lightcurve fit is	3.26 %
Best Comparison Star	#1 - [396, 332]
Optimal Method	PSF photometry
Transit Duration (day)	0.071 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.139 ± 0.043 vs 0.1326 (deviation 0.14σ)
Duration fit vs published	0.071 d vs 0.0737 d (deviation 0.09σ)
Mid-time O-C	-7.4 min
Depth SNR	3.1
Residual scatter	3.26 %

Light curve

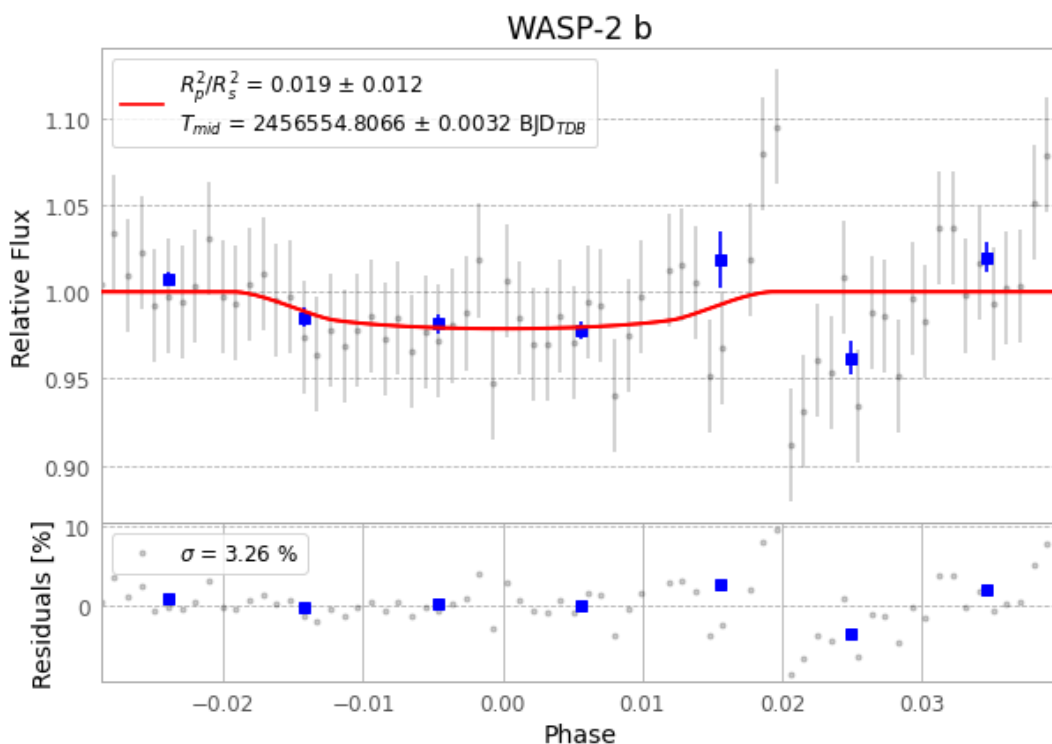


Figure 1 — FinalLightCurve_WASP-2 b_2013-09-18.png (SHA-256 0ee162b54577cac5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [323, 288], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1fe256156da905fa...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-2130919054512.FITS → WASP-2130919093012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-130919.log (8c894cb19434...), FinalLightCurve_WASP-2 b_2013-09-18.png (0ee162b54577...), FinalLightCurve_WASP-2 b_2013-09-18.csv (100b962d9438...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 11:53Z.